Семинар 6

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Термодинамические преобразования.

$$dU_{(S,V)} = TdS - pdV$$

$$d(U - TS) = dF_{(T,V)} = -SdT - pdV$$

$$d(U + pV) = dH = TdS + Vdp$$

$$d(U + pV - TS) = dG = -SdT + Vdp$$

$$\left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial U}{\partial V}\right)_S \qquad \frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}$$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_T ; \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

$$-\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p$$

Гайд на ур-е состояния (универсалочка)

$$dz(x,y) = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy = 0$$

пусть z константа

$$\left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial x}{\partial y}\right)_z = -\left(\frac{\partial z}{\partial y}\right)_x$$

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_z \left(\frac{\partial z}{\partial x}\right)_y = -1$$